SQL PRACTICE

must also learn normalisation

BASIC STRUCTURE OF ANY QUERY

SELECT col1,col2

FROM tableName

WHERE conditions

ORDER BY col_to_sort

LIMIT number of rows;

select specifies the columns we want

from specifies from which table we want to return

ex: select * from emp or select fname,lname from emp

where : how to filter data

returning only rows that meet a specific condition

select * from employees where department = “sales”;

select * from emp where dept in(‘hr’, ‘sales’);

select * from emp where salary between 8000 and 9000;

select * from emp where lname like ‘%n’

select * from emp where salary is null;

ORDERING

list emp salary from highest to lowest

select fname , sal from emp order by sal desc ;

SORTING BY MULTIPLE COLUMNS

select * from emp order by dept asc, sal desc;

so will this query return for each dept salary from highest to lowest like

advertising (high to low),marketing (high to low),sales(high to low)

first pass : dept asc : it sorts all rows based on dept column in ascending alphabetical order , all engineering emp will be grouped together then hr then sales etc

second pass : WITHIN EACH DEPT GROUP ,it then sorts the rows based on the salary column in desc order (highest to lowest)

how it will look like

FirstName	Department	Salary
Charlie	Engineering	95000
Alice	Engineering	90000
Bob	HR	65000
Frank	HR	NULL
Eve	Sales	82000
Diana	Sales	80000

LIMIT

first top highest paid employees

select fname,sal from emp order by sal desc limit 3

Q Find the name and hire date of the two most recently hired emp in the engineering dept

A:

```
select * form fname ,lname , hiredate
from emp
where dept= “Engineering”
order by hiredate desc
limit 2;
```

PRACTICE QUES :

1. Find the FirstName, LastName, and Department of all employees who work in the 'Sales' department.
2. Find all information about employees earning more than $85,000.
3. List the FirstName and HireDate of the 2 oldest employees by hire date (i.e., those hired earliest).
4. Find the EmployeeID and FirstName of any employee whose last name contains the letter 'o'.
5. Find all employees who work in the 'HR' or 'Sales' department and were hired after January 1st, 2022.

   excuse typing mistakes of what I wrote as shortform focus on logic

   correction in bold

1 select fname,lname ,dept from emp where dept= ‘sales’

2 select * from emp where salary > 85000

3 select fname ,hiredate from emp order by hiredate asc **limit 2;**

4 select empid ,fname from emp where lname like ‘%o%’

5 select * from emp where dept in (‘hr’, ‘sales’) and hiredate >1/12022;

correction SQL databases need data to be formatted as strings in specific unambiguous format which is ‘**YYYY-MM-DD’** when we write **1/1/2022** database tries to perform maths calculation (1 divided by 1 divided by 2022) Not what we want

select * from emp where dept in (‘hr’, ‘sales’)

and hiredate > ‘2022-01-01’

Medium Questions:

1. **Find the FirstName, Salary, and Department for all employees who are not in the 'Engineering' department and whose FirstName starts with either 'A' or 'B'. Sort the result by Salary in descending order.**

2. **Find the employee who was the third most recent hire. (Hint: Think about how you can use LIMIT to skip the first two).**
3. **Find all information for employees hired in 2022 whose EmployeeID is an odd number.**

my answers :

1 select fname ,sal ,dept from emp where dept != ‘Engineering’

and fname like ‘A%’ or fname like ‘B%’ order by sal desc

2 select * from employee order by hiredate desc, id desc limit 3

3 select * from emp where year(hiredate)=2022 and mod(empid,2)=1

(or alternate ans can be mod(empid,2)!=0

corrections :

1 And operator has higher precedence than the OR operator ,and database evaluates the and part first

because of this the sql evaluator engine thinks

find emp where

(the dept!=eng AND first name starts with ‘A’)

OR (the first name start with ‘B’)

so It will return an engineer named bob which is not what we want

we must enclose our non AND Logic in parenthesis so that it becomes

select query where 1st condition AND 2nd condtion

select * from emp

dept != ‘engineer’ AND

(fname like ‘A%’ or fname like ‘B%’)

2.Find the employee who was the third most recent hire. (Hint: Think about how you can use LIMIT to skip the first two).

2 . 2nd one only finds the top 3 most recent hires ,but not the 3rd one

we need to use OFFSET : the limit clause has a companion OFFSET .the OFFSET clause skips a specified number of rows before it starts returning rows from limit clause

LIMIT 1 OFFSET m : Skip m rows then get the next 1 row

```
select * from emp
order by hiredate desc
limit 1 offset 2
```

```
select * from emp
order by hiredate desc
offset 2 rows
fetch next 1 rows only
```

3.Find all information for employees hired in 2022 whose EmployeeID is an odd number.

```
select * from emp
where extract(year from hiredate)=2022
and empid%2!=0
```

BLOCL 1.5 sql funcs

concat () : join 2 or more string together

ex concat (fname , ‘ ’,lname)

lower() / upper() : convert string to all lowercase or all uppercase

ex where lower(fname)= ‘alice’

length() /len() : returns number of chars in string

length(‘hello’) will return 5

substring()/substr() : extract portion of string

---

q1 average salary of all employees

select coalesce(avg(salary),0) from employees;

q2 Find the number of employees working in each department. The result should show the department name and the employee count.

select department,count(empid)

as NoOfEmployees

from employees

group by department

q3For each department, find the salary of the highest-paid employee.

select department , max(salary) as highestSalary

```
from employees
group by department
```

q4 List the departments where the average salary is less than $90,000.

```
select department ,avg(salary)
from employee
group by department
having avg(salary)<90000
```

q5 Find the total salary expense for each department. Only show departments with a total salary expense greater than $150,000. Sort the result from highest to lowest total expense.

```
select department,sum(salary)
from employees
group by department
having sum(salary)>150000
order by desc;
```

```
select department,sum(salary)
as totalSalaryExpense
from employees
group by department
having sum(salary)>150000
```

order by totalSalaryExpense desc;

Find the earliest and the most recent hire dates across the entire company.

select min(hiredate),max(hiredate) from employees

How many employees in each department were hired before the year 2023?

select department,count(empid) as empcount

from employees

where hiredate<'2023-01-01'

group by department

For each department, calculate the average salary. Only include employees who actually have a salary on record.

select department , avg(salary) as averageSalary

from employees

group by department;

List the departments where the highest salary is exactly $95,000.

```
select department,max(salary)
from employees
group by department
having max(salary)=95000
```

Find the number of employees for each manager. Show the ManagerID and the count of employees who report to them.

```
select count(empid),managerId
from employees
group by managerid
```

From the previous result, only show the managers who have more than one direct report.

```
select count(empid),managerId
from employees
group by managerid
having count(empid)>1
```

Calculate the salary range (the difference between the maximum and minimum salary) for the 'Engineering' department. Name the result column SalaryRange.

```
select (max(salary)-min(salary))
as SalaryRange
from employees
where department='Engineering'
```

Count the number of employees hired in each year. The result should show the year and the number of employees hired in that year.

```
select count(empid),hiredate
from employees
group by hiredate
```

```
select year(hiredate) as hireyear,count(empid)
from employees
group by hireyear;
```

JOINS

Table: Employees

EmployeeID	FirstName	DepartmentID
101	Alice	1
102	Bob	2
103	Charlie	1

104 Diana 3

Table: Departments

DeptID	DeptName
1	Engineering
2	HR
4	Marketing

inner join most common

join condition is met on both tables

```
select Employees.firtsname,
departments.deptid
from employees
inner join department
on employees.deptid=dept.deptid
```

ALSO DO FOR THREE TABLES JOIN FOR THREE TABLES

Questions:

(INNER JOIN) List the Name of each customer and the Amount of every order they placed.

```
select c.name,o.amount
from customers c
inner join orders O
on c.customerID=o.customerID
```

(LEFT JOIN) Show a list of all customer names. If a customer has placed an order, also show the OrderID.

```
select c.name,o.orderid
from customer c
left join order o
on c.customerid=o.customerid
```

(LEFT JOIN with WHERE) Find the names of customers who have not placed any orders. (This is a very common and important interview question!)

```
select c.name
from customer c
left join order o
on c.customerid=o.customerid
where orderid IS NULL
```

not orderid=null since null is a state and not a vlaue

(JOIN with GROUP BY)

Calculate the total amount spent by each customer. Show the customer's Name and their total spending, sorted from highest to lowest.

must group by cname and not amount !!

```
select c.name,sum(o.amount)
as totalSpending
from customers c
inner join orders o
on c.customerId=o.customerid
group by c.name
order by totalSpending desc;
```